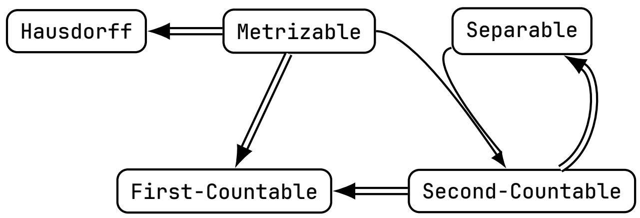
0.0.1 Metrizable Theorem Diagram



Topology for $\mathbb{R}$: $\mathcal{T}_{id} \subset \mathcal{T}_f \subset \mathcal{T}_u \subset \mathcal{T}_l \subset \mathcal{T}_d,\ \mathcal{T}_f \subset \mathcal{T}_c$						
Topology	$\mathcal{T}_{id}$	$\mathcal{T}_f$	$\mathcal{T}_u$	$\mathcal{T}_l$	$\mathcal{T}_d$	$\mathcal{T}_c$
First-Countable	Yes	No	Yes	Yes	Yes	No
Second-Countable	Yes	No	Yes	No	No	No
Hausdorff	No	No	Yes	Yes	Yes	No
Metrizable	No	No	Yes	No	Yes	No
Separable	Yes	Yes	Yes	Yes	No	No
Compactness	Yes	Yes	No	No	No	No

Let a Topological Space $(X, \mathcal{T})$ be given. For X, the following theorems hold:

Theorem 0.0.1. If Metrizable, then Hausdorff.

Proof. Let $(X, \mathcal{T})$ be a metrizable topological space. Then, there is a metric $d : X \times X \to \mathbb{R}$.
Let distinct two point $x, y \in X$ be given. And, put $\varepsilon = \frac{d(x,y)}{2} > 0$.
Now, $x \in B_d(x; \varepsilon)$ and $y \in B_d(y; \varepsilon)$ but there intersection is empty set because:
Choose $p \in B_d(x; \varepsilon) \cap B_d(y; \varepsilon)$. Then, $d(x, p), d(y, p) < \varepsilon$.
But, this implies that

$$d(x, y) \leq d(x, p) + d(p, y) < 2\varepsilon = d(x, y)$$

Thus, contradiction. Consequently, given topological space satisfies Hausdorff property. □

Lemma 0.0.2. Let (X, d) be a Metric Space.
For any $U \in \mathcal{T}_d$, $x \in U$, there is $\delta > 0$ such that $x \in B_d(x, \delta) \subset U$.

Proof. Let $U \in \mathcal{T}_d$ be given.
Then, there exist $y \in X$ and $\varepsilon > 0$ such that $x \in B_d(y, \varepsilon) \subset U$. For $\delta = \varepsilon - d(x, y) > 0$, $B_d(x, \delta) \subset B_d(y, \varepsilon)$ because:

$$\begin{aligned} p \in B_d(x, \delta) &\iff d(p, x) < \delta \\ &\implies d(p, y) \leq d(p, x) + d(x, y) < \delta + d(x, y) = \varepsilon \\ &\iff p \in B_d(y, \varepsilon) \end{aligned}$$

□

Theorem 0.0.3. If Metrizable, then First-Countable.

Proof. Let $(X, \mathcal{T})$ be a metrizable topological space, and d be a metric that induces $\mathcal{T}$.
For any $x \in X$, define

$$\beta_x \overset{\text{def}}{=} \left\{ B_{\frac{1}{n}}(x) \,\middle|\, n \in \mathbb{N} \right\}$$

Trivially, β_x is countable set. Want To Show: β_x be a local base of x.
LB1) is clear.
LB2) Let $U \in \mathcal{T}$ with $x \in U$ be given. Since above lemma, there is $\delta > 0$ such that $x \in B_d(x, \delta) \subset U$.
Let $n \in \mathbb{N}$ be a number such that $1 < \delta n$, then $B_d\left(x, \frac{1}{n}\right) \in \beta_x$ satisfies

$$x \in B_d\left(x, \frac{1}{n}\right) \subset B_d(x, \delta) \subset U$$

□

> **Theorem 0.0.4.** If Metrizable and Separable, then Second-Countable.

Proof. Since X is Separable, let $A = \{x_i \mid i \in \mathbb{N}\}$ be a countable dense subset of X.
Let $d : X \times X \to \mathbb{R}$ be a metric on X that induces $\mathcal{T}$. Define

$$\beta \overset{\text{def}}{=} \left\{ B_d\left(x_i, \frac{1}{n}\right) \,\middle|\, i, n \in \mathbb{N} \right\}$$

Clearly, β is Countable and $\beta \subset \mathcal{T}$.
Enough to Show that: For any $U \in \mathcal{T}$, there exists $\mathcal{C} \subset \beta$ such that

$$U = \bigcup_{u \in \mathcal{C}} u$$

Let $U \in \mathcal{T}$ be given. Since $\mathcal{T}$ is consisted by open balls induced a metric d,
for any $x \in U$, there exists $r > 0$ such that $B_d(x, r) \subset U$ by the lemma.
Set $\delta_x = \frac{1}{2n}$ such that $1 < nr$. Since A is dense in X,

$$A \cap B_d\left(x, \delta_x\right) \neq \emptyset$$

That is, for some $i \in \mathbb{N}$, $d(x, x_i) < \delta_x$. Now,

$$x \in B_d\left(x_i, \delta_x\right) \subset B_d(x, 2\delta_x) \subset B_d(x, r) \subset U$$

Consequently, we shown that:

$$\forall x \in U,\ \exists B_x \in \beta \ \text{ s.t } \ x \in B_x \subset U$$

Now, Construct given $U \in \mathcal{T}$ as: Since **Axiom of Choice**, Define $\mathcal{C} \overset{\text{def}}{=} \bigcup_{x \in U} \{B_x \in \beta \mid x \in B_x \subset U\}$. Then,

$$U = \bigcup_{u \in \mathcal{C}} u$$

□

> **Theorem 0.0.5.** If Second-Countable, then Separable.

Proof. Let $\beta = \{B_i \mid i \in \mathbb{N},\ B_i \neq \emptyset\}$ be a Countable Basis of X.
By the **Axiom of Choice**, we can construct a countable set $A = \{x_i \mid i \in \mathbb{N},\ x_i \in B_i\}$. WTS: $\bar{A} \supset X$.
Let $x \in X$ and $U \in \mathcal{T}$ with $x \in U$ be given arbitrarily.
Since β is basis of $\mathcal{T}$, $x \in B_i \subset U$ for some $B_i \in \beta$. Thus, $x_i \in U$, so we obtain $A \cap U \neq \emptyset$. Therefore, $x \in \bar{A}$. □

> **Theorem 0.0.6.** If Second-Countable, then First-Countable.

Proof. Trivial. □